

HumanEd: "Oar" Robot

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1 Overview



Figure 1: Mockup of Oar

The "Oar" robot is a 3D printed humanoid robot comprised of a rigidly-supported torso, a head placed upon an actuated neck, and two 7 degree of freedom robotic arms. A basic mockup of Oar is shown in Figure 1.

The design goals for Oar are as follows:

- Demonstrate the strengths, capabilities, and pitfalls of 3D printed actuator gearhead designs and serve as a design study that may inform future HumanEd humanoid robots.
- Provide a reliable humanoid arm system capable of accurately positioning the end-effector for experimental and demonstratory purposes. Experimental usages includes testing of end-effector designs (grippers, hands), a real-world interface for computer vision experiments, and a follower-arm for teleoperations experiments. Demonstrations may include various tasks that could be performed at fairs or open days at which the society of HumanEd may be present.
- Provide a reasonable suite of sensors that can be interfaced with for experiments. Such experiments may include computer vision or reinforcement learning projects.

This proposal will focus primarily on the development of the robot's arms as they are considered to be the primary engineering challenge, however some notes on the body, neck/head, and end-effector will still be made.

1.1 A note on the name

The name "Oar" has been derived from the first line of the poem "Basking Shark" by Scottish poet Norman MacCaig [1], which reads:

*To stub an **oar** on a rock where none should be*

The poem is a discussion on themes of the origins of life and the place of humanity in the timeline of evolution. The decision to choose this poem to name the robot reflects the robot's intended position as HumanEd's first full-scale humanoid robot. It is also intended for an inscription of the poem to be placed somewhere on the robot in commemoration of this poem.

2 Arm Design

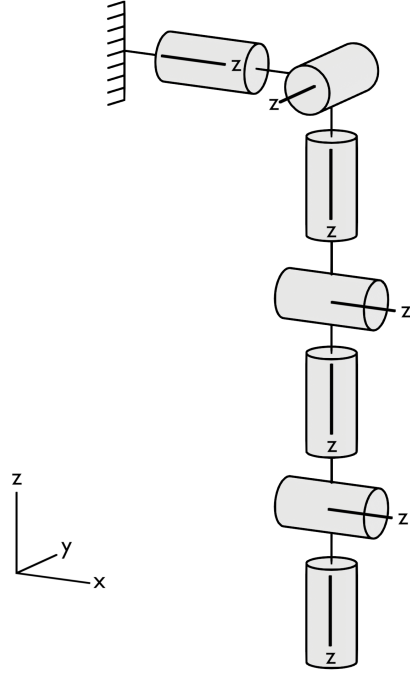


Figure 2: Joint diagram of Oar's left arm

The arms of Oar will follow the common 7-DOF humanoid arm model of a 3-DOF shoulder and a 3-DOF wrist connected by a 1-DOF elbow. Figure 2 shows the joint diagram of the left arm.

The arm aims to have a reach of 600 mm (measured from the axis of rotation of the shoulder pitch actuator to the center of the gripper's jaws). To keep costs down, the arm's actuators will be driven using stepper motors. The majority of the mechanical and aesthetic parts will be FDM printed, with the major exceptions of bearings, fixtures, and some elements with specific material requirements (such as the usage of aluminium extrusion to resist high bending moments, or of steel dowels to reduce surface frictions as in § 3.3).

The target payload is 500 g, this has been chosen based on the PAROL6 robot by *Source Robotics* [2], which is a 6-DOF robotic arm that also makes heavy usage of FDM parts and is driven by stepper motors, and has a maximum reach of 400 mm and payload of 1 kg.

2.1 Forward Kinematics & Preliminary Torque Analysis

All of the joints shown in Figure 2 are revolute joints. For the purpose of the forward kinematics analysis, all three of the shoulder joints will be considered coincident, and all three of the wrist joints will also be considered coincident. The vertical distance between the shoulder joints and the elbow joint is taken as L_1 , and the distance between the elbow joint and the wrist joints as L_2 .

The resting position of the end-effector frame $\{b\}$ is a right-handed set of axes rotated by π radians about the x-axis such that the z_b axis points downwards.

The forward kinematics of the arm can then be defined via the method as described by Lynch and Park [3, § 4.1]. $\theta \in \mathbb{R}^7$ is the vector of all joint positions of the arm, and $M \in SE(3)$ is the

transformation matrix of the end-effector when the arm is in its resting position, defined as

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & -(L_1 + L_2) \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3)$$

The screw axes $\mathcal{S}_i$, $i = 1, \dots, 7$, represented in the space frame, can be seen as:

$$\begin{aligned} \mathcal{S}_1 &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{S}_4 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -L_1 \\ 0 \end{bmatrix} \\ \mathcal{S}_5 &= \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}, \quad \mathcal{S}_6 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}, \quad \mathcal{S}_7 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix} \end{aligned}$$

Which yields the forward kinematics equation where $T(\theta) \in SE(3)$ is the transformation matrix describing the position and orientation of the end-effector given the joint positions θ :

$$T(\theta) = {}_e[\mathcal{S}_1]\theta_1 {}_e[\mathcal{S}_2]\theta_2 {}_e[\mathcal{S}_3]\theta_3 {}_e[\mathcal{S}_4]\theta_4 {}_e[\mathcal{S}_5]\theta_5 {}_e[\mathcal{S}_6]\theta_6 {}_e[\mathcal{S}_7]\theta_7 M \quad (1)$$

Equation 1 can then be used to find the space Jacobian $J_s(\theta) \in \mathbb{R}^{6 \times 6}$ of the arm's end-effector via the expression:

$$J_{si}(\theta) = \left[\text{Ad}_{{}_e[\mathcal{S}_1]\theta_1 \dots {}_e[\mathcal{S}_{i-1}]\theta_{i-1}} {}_e[\mathcal{S}_i]\theta_i \right] \mathcal{S}_i \quad (2)$$

where $J_{si}(\theta)$ is the i th columns of $J_s(\theta)$. This can then be used to perform a statics analysis of the arm via the equation

$$\tau = J_s^T(\theta) \mathcal{F}_s \quad (3)$$

where $\mathcal{F}_s$ is a wrench applied to the end-effector, and τ is the resulting vector of torques required in each joint to resist $\mathcal{F}_s$.

The method for performing this analysis is detailed in [3, § 5] and for this robot results in matrices that are too large and complex to be reasonably computed by hand for every possible configuration of interest. As such Equations, 1, 2, and 3 were implemented into a python script to complete the analysis.

Using the python script, a variety of configurations were tested for a 1 kg payload. The tested configurations showed that the maximum torque required of the arm's joints would take place in the shoulder pitch, shoulder roll, and elbow, and would be roughly of a magnitude of 6 Nm. This theoretical payload is double the target discussed in § 1 as the arm must also be able to lift its own weight.

This result was then used to inform the design of various prototype gearheads. Considering an estimated dynamic torque of 0.2 Nm for a mid-sized NEMA17 stepper motor, the calculated torque figure of 6 Nm, and some additional factor of safety, a target reduction ratio of 48:1 was chosen for the prototype gearheads. The development of the prototypes are detailed in § 3.

2.2 End-Effector Design

In order to keep the mass low, the initial end-effector will be a simple gripper driven using a servo, converting its rotational output into two linearly moving jaws. This style of gripper can be seen on the AR4 robotic arm by *Annin Robotics* [4].

A more advanced end-effector may be a pneumatic gripper or one with maleable jaws to allow for the handling of delicate objects.

To allow for efficient experimentation, the end-effectors should be easy to detach and reattach without any major dissassembly of the wrist mechanism.

3 Gearhead Prototypes

Cycloidal drives have been chosen for the actuators of the arm. This is primarily due to their affinity for being produced using FDM printing when compared to planetary or strain-wave gear systems, wherein the former presents difficult to achieve tolerances (as well as comparatively low reduction ratios) and the latter requires the development of a flexible element.

The following sections detail the development of three prototype gearheads. The model of stepper motor used for each prototype was the 17HS3401S, which was ran at a speed of 100 RPM and measured to have a dynamic torque of 0.17 Nm¹. The cycloidal discs printed for each prototype followed the 0.3 mm tolerance determined by Quazi [5].

The output torque produced by each prototype gearhead was tested using various masses suspended from a 30 cm bar of 20 mm $\times$ 40 mm aluminium extrusion attached from the centerpoint of the gearhead’s output.

Table 1 shows a summary of the results for all three prototypes.

No.	Reduction Ratio	Mass (g)	Maximum Measured Output Torque (Nm)	Efficiency
1	24:1	420	1.0	25%
2	48:1	460	1.4	17%
3	48:1	740	3.1	38%

Table 1: Summary of prototype gearheads.

3.1 First Prototype - 24:1

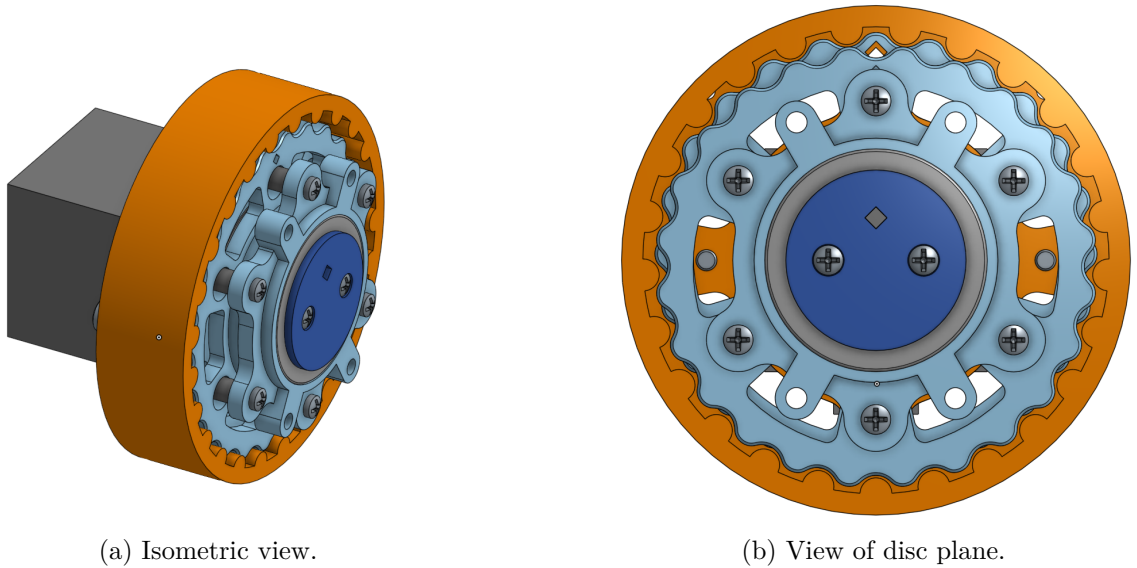


Figure 3: The prototype 24:1 gearhead design.

Figure 3 shows the design of the first prototype gearhead that was built and tested. This prototype was built with a 24:1 reduction ratio, half of the primary 48:1 reduction ratio calculated in § 2.1. Figure 3a shows an isometric view of the design with all housings included, Figure 3b shows a view from the plane of the gearhead’s cycloidal discs, with the output housing hidden for clarity.

The design is almost entirely comprised of FDM printed parts, with the only exceptions being the bearings, screws, and of course stepper motor.

¹Used to drive a rudimentary pulley of diameter 30 mm and found to be capable of lifting a maximum of 575 g.

The gearhead was seen to be able to achieve a maximum output torque of roughly 1.0 Nm, which gives an efficiency of about 25%. This is, needless to say, a poor result, however it did show that the design was able to successfully increase the output torque of the motor.

3.2 Second Prototype - 48:1

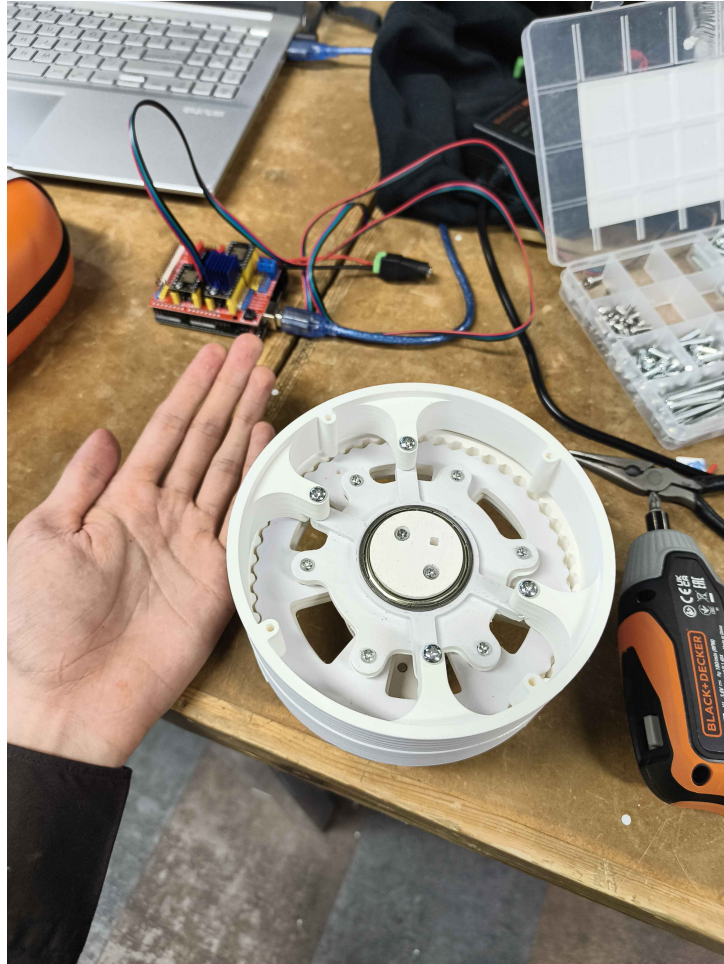


Figure 4: A photo of the second constructed prototype gearhead, hand for scale.

The primary cause of the low efficiency seen with the first prototype in § 3.1 was believed to be excess friction caused by poor tolerances and printed part surface finishes. The second prototype, shown in Figure 4, was produced using a higher quality printer in order to improve these tolerances and surface finishes. Due to added material availability, it was also decided to scale the prototype up to the full 48:1 design. The design of this second prototype was fundamentally the same as the first, only scaled up.

Despite the improved print quality, efficiency actually decreased from the initial prototype. The highest output torque achieved was roughly 1.4 Nm, which gives an efficiency of 17%.

3.3 Third Prototype - 48:1 with Steel Dowels

Following the low efficiencies of the first and second prototypes, as seen in § 3.1 and § 3.2 respectively, it became clear that the frictional forces present between the rough surfaces of FDM printed parts were too high to allow an effective gearhead to be produced without the usage of some lower frictional elements.

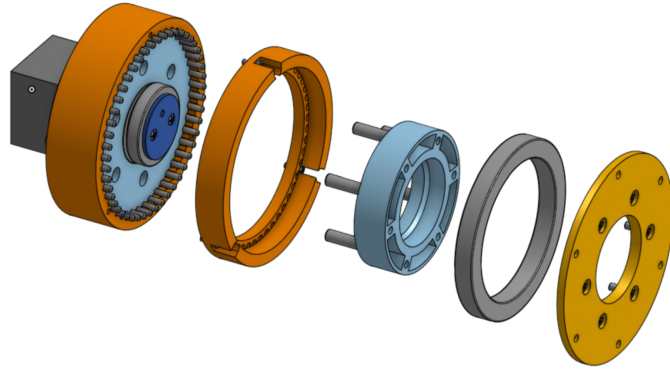


Figure 5: A partially exploded view of the third prototype gearhead.

As such, the third prototype gearhead, as seen in Figure 5, made use of steel dowels for both the inner and outer pins of the cycloidal drive. An additional large diameter bearing was also added to stabilise the output disc and prevent friction from its lateral warping, which was observed in the prior prototypes. PTFE synthetic grease was also applied as a lubricant to all surfaces subject to high friction.

This prototype was seen to achieve a maximum output torque of 3.1 Nm, giving an efficiency of 38%.

4 Body, Head, & Sensors

4.1 Body

As the focus of Oar is its arms, the body would simply be some portion of an upper torso, sized to fit the necessary electronics, and mounted on a rigid stand of sensible construction. On the back face of the body would also be housed all the external connections needed to use the robot, such as power and USB ports.

Wheels may be considered for the base of the stand in order to make transporting the robot easier. It would additionally be helpful for the stand to be adjustable in height. A battery may also be considered for placement within the body or at the base of the stand as the robot is already likely to be top-heavy, however as the robot itself is only intended for operation when placed stationary, this is not of immediate concern. As such, the robot can simply be powered via electrical mains and a suitable power supply.

While independent power is not a major concern, possessing a suitable controller is an important element that must be fitted within the body. A Raspberry Pi or other suitable robotics board that is capable of running ROS should be implemented into the body.

Depending on the hardware chosen, cooling fans may also be a sensible addition in order to safely manage the heat generated via the running of the robot's many stepper motors.

4.2 Head

The head is to be mounted upon an actuator at the center of the top surface of the body. The most basic construction for the neck would be a single revolute joint, allowing the head to pivot about a vertical axis. A more advanced neck would be constructed of two revolute joints in order to allow for spherical motion of the head. The latter design would be preferable in order to allow for the widest range of capabilities in experiments (such as in computer vision experiments), however depending on the sensor hardware fitted to the head, may be an unnecessary amount of freedom of motion.

4.3 Sensors

Joint rotational encoders are mentioned in § 5.

In order to be able to interact with its environment, Oar must possess various sensors.

The most obvious option is a camera placed within the head. This would facilitate computer vision and potentially remote operation for teleoperations experiments. An accompanying infrared sensor to measure distances would provide useful assistance in manipulation experiments that require Oar to determine the position of objects.

Touch or force sensors may be implemented into Oar's grippers or any other type of relevant end-effector to aid in tasks requiring delicate grasping.

Although a common fixture on many robots, as Oar is ultimately a stationary robot, an inertial measurement unit is not necessary to be implemented. However, if testing shows it to be worthwhile, gyroscopes placed within the end-effectors may serve as useful calibration or corrective sensors.

5 Steps for Development

This section details the next steps that would occur in the mechanical development of Oar’s arms.

5.1 Further Gearhead Prototypes

The 38% maximum efficiency achieved by the third gearhead prototype seen in § 3.3 is considerably lower than the efficiency figures reported by other designs, which range from 60% [6] to over 90% [7].

One potential reason for this is that the designs by Nedelkovski [6] and *Advanced Hobby Lab* [7] featured significantly lower reduction ratios of 15:1 and 8:1 respectively. It is possible that the losses seen in the prototype 24:1 and 48:1 gearheads were due to the greatly increased surface area of their cycloids. Furthermore, Nedelkovski’s design utilises rolling elements wherever possible to reduce losses to friction.

For these reasons, a logical next step would be to develop two similar low ratio gearheads, the first of which will use rolling elements for all inner and outer pins in order to maximise efficiency. This gearhead would then be used to develop the gearheads present in the shoulder and elbow actuators, which are the joints the arm required to output the highest torques. The high ratios required for these joints can be obtained by developing two-stage gearheads (for example, combining two 10:1 cycloidal designs to produce a 100:1 reduction ratio).

From this prototype, a second gearhead will be developed with the intent of minimising weight. Heavy components such as bearings or steel dowels can be strategically swapped out for FDM parts in order to balance efficiency with mass. This will be used to develop the gearheads of the wrist joints, which will be subjected to significantly smaller moments than the upper arm joints.

Additionally, these new gearhead prototypes should be implement rotational encoders as these will be important in safely controlling Oar.

5.2 Tip to Base Design Order

Once satisfactory gearhead prototypes have been developed, work on the full arm can begin. The arm should be developed from the end-effector to the shoulder, such that the gearheads and structural design of each subsequent section can be informed by the weight of the previous section.

Particular attention should be paid to the end-effector and the wrist joints, with efforts placed on making them as light as possible. In light of this goal, pancake-style NEMA stepper motors can be considered for these joints.

If necessary, belts transmissions can be considered in order to bring heavy actuators closer to the shoulder in order to reduce the magnitude of the moments they produce. In particular this can be done for the elbow, which can have its gearhead placed directly after the shoulder and the elbow joint itself actuated via belt. This may also be possible for the wrist pitch, however depending on the size of the gearheads involved, the distance which the gearhead may be able to be moved back up the arm may not be enough to reduce its moment enough to justify the added complexity. A belt transmission solution would also require encoders to be placed at the joints themselves rather than on the actuators.

For the shoulder joints, particularly the shoulder pitch and shoulder roll, higher torque motors can be considered to drive their actuators. This could simply be a larger NEMA17, or for the shoulder pitch (which will be mounted within Oar’s body and thus will not create any moments on the arm) potentially even a NEMA23 stepper motor.

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